

MALAGA AP, Spain

WMO: 084820

Lat: 36.666N

Lon: 4.482W

Elev: 52

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	39.6	42.5	23.4	17.7	52.8	28.2	22.2	53.8	26.9	57.1	24.9	58.0	10.0	300	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.4	96.1	68.2	91.8	68.2	88.2	68.2	75.6	83.6	74.5	82.5	73.5	81.3	11.2	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.1	123.3	80.6	71.9	117.9	79.8	70.8	113.5	79.1	39.0	84.1	38.0	82.5	37.0	81.4	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	20.2	18.0	DB	35.2	103.2	2.3	2.9	33.6	105.4	32.2	107.1	30.9	108.7	29.2	110.9
			WB	32.3	77.1	2.7	1.8	30.3	78.4	28.8	79.5	27.3	80.5	25.3	81.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	65.9	54.7	55.6	58.5	62.1	67.4	74.2	78.7	79.6	74.5	68.0	60.4	56.1
	DBStd	9.80	4.24	4.23	4.17	4.22	4.56	4.35	4.35	4.00	4.31	4.25	4.95	4.01
	HDD50	19	9	5	1	0	0	0	0	0	0	0	1	3
	HDD65	1373	320	263	205	108	26	0	0	0	1	19	154	277
	CDD50	5812	154	162	266	362	538	725	889	919	736	558	312	191
	CDD65	1692	0	1	5	20	99	275	424	454	287	112	15	0
	CDH74	13893	1	4	35	117	561	2092	4110	4526	1993	417	37	0
	CDH80	4804	0	0	4	15	123	657	1622	1777	542	62	2	0
Wind	WSAvg	8.1	9.5	9.2	8.5	8.1	7.8	7.8	7.6	7.3	6.9	6.8	8.5	9.0
Precipitation	PrecAvg	20.5	2.4	2.3	2.3	1.7	1.0	0.3	0.0	0.1	1.1	2.4	3.1	3.5
	PrecMax	43.0	12.1	8.8	9.8	5.7	4.4	3.2	0.2	0.7	5.6	7.6	8.8	12.8
	PrecMin	11.3	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0
	PrecStd	8.1	2.8	2.0	2.0	1.3	1.1	0.6	0.0	0.2	1.3	1.8	2.2	3.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.4	73.6	78.4	82.0	88.2	95.8	101.8	101.0	94.5	86.4	78.5	70.7
		MCWB	56.8	57.9	59.1	62.1	64.8	67.6	69.1	69.0	68.0	66.6	62.0	58.9
	2%	DB	67.2	69.4	73.2	77.1	83.5	90.7	96.6	95.4	89.0	80.7	73.8	67.9
		MCWB	56.0	56.7	58.1	60.1	63.0	66.8	68.3	68.6	67.7	65.9	61.4	58.0
	5%	DB	64.6	66.2	70.0	73.8	79.5	86.4	91.7	91.6	85.5	78.4	71.2	65.7
		MCWB	54.9	55.6	57.0	59.2	62.3	66.0	67.7	69.3	68.1	66.3	60.8	57.0
	10%	DB	62.5	63.9	67.1	71.4	76.7	83.6	87.7	87.8	82.6	75.6	68.4	63.5
		MCWB	54.3	54.6	56.5	58.4	61.8	65.9	68.7	69.7	68.8	65.2	59.4	56.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.2	60.9	62.7	65.3	68.3	73.1	77.1	77.3	75.3	70.8	66.5	62.1
		MCDB	66.4	69.3	71.0	75.2	78.2	82.7	84.4	85.6	83.4	78.8	73.3	66.9
	2%	WB	58.0	58.8	60.4	63.2	66.6	71.4	75.0	76.1	73.8	69.4	64.2	60.1
		MCDB	64.1	65.9	68.8	71.8	76.2	80.8	83.0	84.1	81.1	76.6	70.6	64.9
	5%	WB	56.6	57.3	59.1	61.6	65.3	70.0	73.8	75.1	72.5	68.3	62.7	58.6
		MCDB	62.4	63.4	66.7	69.9	74.5	79.5	81.7	83.1	79.8	74.8	68.4	63.5
	10%	WB	55.2	55.9	57.9	60.2	64.0	68.9	72.7	74.1	71.2	67.0	61.1	57.2
		MCDB	61.0	61.9	64.9	68.3	72.8	78.4	81.0	82.1	78.7	73.2	66.7	62.0
Mean Daily Temperature Range	5% DB	MDBR	15.7	16.1	16.4	16.9	16.9	17.0	17.2	16.4	15.7	15.1	15.0	14.7
		MCDBR	17.8	19.4	20.8	21.0	21.0	21.8	24.3	22.7	20.0	17.8	16.8	16.5
		MCWBR	10.9	11.5	11.3	10.5	9.5	9.6	10.3	10.5	9.8	9.2	9.5	10.1
	5% WB	MCDBR	16.4	17.5	17.7	17.8	17.1	16.9	15.8	14.7	15.2	14.6	14.0	14.4
		MCWBR	11.4	11.8	11.2	10.8	9.8	9.8	9.6	9.5	9.5	9.1	9.4	10.1
Clear-Sky Solar Irradiance	taub	0.321	0.347	0.385	0.390	0.407	0.429	0.426	0.435	0.413	0.383	0.345	0.326	
	taud	2.528	2.467	2.355	2.359	2.341	2.292	2.295	2.274	2.364	2.455	2.522	2.536	
	Ebn at Noon	277	282	279	283	279	271	271	266	266	265	266	267	
	Edh at Noon	25	30	37	38	40	42	42	42	36	30	25	23	
All-Sky Solar Radiation	RadAvg	860	1112	1495	1872	2218	2504	2495	2204	1721	1242	903	744	
	RadStd	73	127	158	129	148	72	56	66	96	102	77	75	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.79	N/A	N/A	N/A	+0.65	N/A	N/A	N/A	+304	+225
(47)	Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+113

Nomenclature: See separate page